



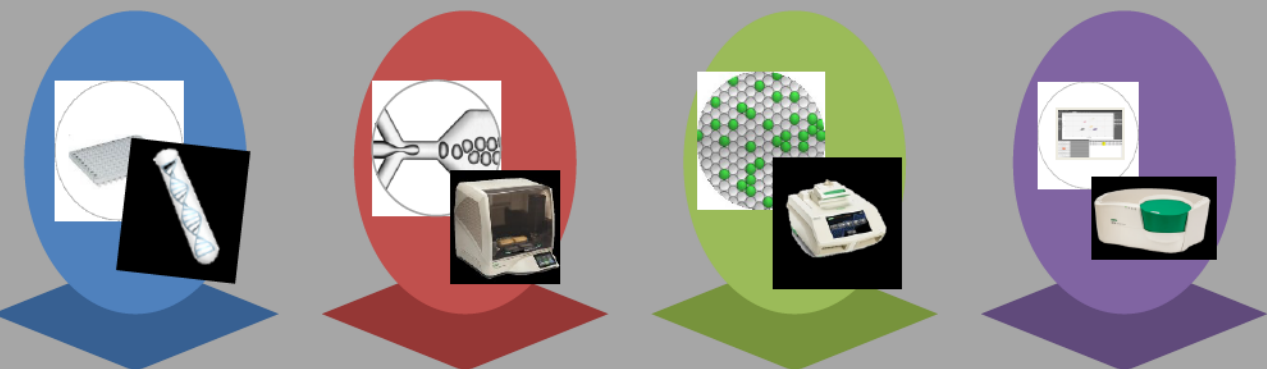
VeriCheck ddPCR *Mycoplasma* Detection Kit: Probe-based *Mycoplasma* Detection to Reduce False-Positives Results

BIO-RAD

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VeriCheck ddPCR *Mycoplasma* Detection solution

The VeriCheck ddPCR *Mycoplasma* Detection Kit is the first Droplet Digital PCR (ddPCR) based *Mycoplasma* testing solution. The **probe-based, sensitive, and specific kit** has been designed and validated to meet European, U.S., and Japanese pharmacopoeia requirements and can detect up to **112 *Mycoplasma* species**. Utilize the U.S. FDA 21 CFR Part 11 compliant software and **positive-control based autothresholding** for turnkey, easy data analysis.



Prepare Reaction Generate Droplets Perform PCR Read and Analyze

Combine assay, ddPCR supermix, and DNA samples or controls ~20,000 droplets per well, automated One-step protocol for endpoint amplification Easy-to-use software with integrated controls and flexible tools

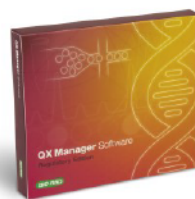
The VeriCheck ddPCR *Mycoplasma* Detection Kit provides:

- Automated data analysis able to calculate copies/well without user intervention
- A Limit of Detection (LOD) of ≤ 1 colony forming units (CFU)/ml, and ≤ 6 Genome Copies (GC)/well
- Specificity to *Mycoplasma* species and not to related bacteria
- Tolerance to common PCR inhibitors



Easy Analysis: QX Manager Regulatory Edition v1.2 Enables Control-Based Autothresholding

VeriCheck ddPCR *Mycoplasma* detection Kit has been validated for automated data analysis based on the positive control. The automated data analysis is part of the accompanying U.S. FDA 21 CFR Part 11 compatible QX Manager Software, Regulatory Edition. The software also offers audit trails with tracked protocol changes for regulatory requirements.



Sensitivity: Limit of Detection (LOD) measured in colony forming units (CFU) per well

Concentration	Species Name	Kit Lot 1		Kit Lot 2	
		n	Concentration (Copies/well)	n	Concentration (Copies/well)
1 CFU/mL	<i>Acholeplasma laidlawii</i>	4	18.8	4	17.8
	<i>Mycoplasma pneumoniae</i>	3	11.5	17	9.9
	<i>Mycoplasma gallisepticum</i>	4	7.2	4	10.0
	<i>Mycoplasma hyorhinis</i>	4	5.5	4	4.6
	<i>Mycoplasma orale</i>	4	5.6	4	13.6
	<i>Mycoplasma fermentans</i>	4	3.1	4	5.7
	<i>Mycoplasmasynoviae</i>	4	6.6	4	5.2
	<i>Mycoplasma arginini</i>	4	31.5	4	33.9
	<i>Mycoplasma salivarium</i>	4	10.3	4	10.8

Table 1. Detection and quantification of each of 9 mycoplasma species at 1 CFU/mL. All species were detected at 1 CFU/ml at $\geq 95\%$ positive.

Two lots of the VeriCheck ddPCR *Mycoplasma* Detection Kit were tested for 9 species extracted at 1 CFU/ml. Each plate contained three replicates of each control and four replicates of each sample. For the nine *Mycoplasma* species, $\geq 95\%$ of the replicate wells were positive for *Mycoplasma* extracted at 1 CFU/ml. The LOD is 1 CFU/ml

Sensitivity: Limit of Detection (LOD) measured in genome copies (GC) per well

Species	Lot #	LOD (Expected GC/well)	LOD (Observed GC/well)
<i>A. laidlawii</i>	1	3	3.03
<i>M. hyorhinis</i>	1	5	3.71
<i>M. pneumoniae</i>	1	5	5.25
<i>A. laidlawii</i>	2	5	3.54
<i>M. hyorhinis</i>	2	3	2.78
<i>M. pneumoniae</i>	2	10	6.29
<i>A. laidlawii</i>	3	5	4.19
<i>M. hyorhinis</i>	3	5	5.63
<i>M. pneumoniae</i>	3	3	3.61
<i>A. laidlawii</i>	1	3	2.95
<i>M. hyorhinis</i>	1	3	2.65
<i>M. pneumoniae</i>	1	3	3.17

Table 2: The measured GC/well for three representative mycoplasma species tested with three kit lots, on both QX200 and QX ONE ddPCR systems. The requirement for LOD measurement is $\geq 95\%$ positive. $n > 100$ for each species. LOD ≥ 6 GC/well.

The three *Mycoplasma* species *A. laidlawii*, *M. pneumoniae*, and *M. hyorhinis*, that represent all nine mycoplasma species, were used in this study. ddPCR testing of each standard was carried out by three operators using three different development lots on the QX200 ddPCR system, and one operator using one lot on the QX ONE ddPCR system. The LOD of *A. laidlawii* is 4.19 GC/well, *M. pneumoniae* is 6.29 GC/well, and *M. hyorhinis* is 5.63 GC/well, the three representative species for all 9 mycoplasma species. Similar performance was seen on the QX200 and QXONE platforms. The LOD is ≤ 6 GC/well.

Specificity: Kit does not detect *S. bovis*, *L. acidophilus*, and *C. sporogenes* per European Pharmacopoeia recommendation

Genomic DNA (gDNA) samples from the three representative *Mycoplasma* species, *A. laidlawii*, *M. pneumoniae*, *M. hyorhinis*, and the no-detect species *C. sporogenes*, *L. acidophilus*, and *S. bovis*, were obtained from ATCC and Minerva Biolabs. The contrived samples were tested with the VeriCheck ddPCR *Mycoplasma* Detection Kit for any cross-reactivity, using the QX200 ddPCR System. The results of this experiment (figure 1) demonstrate the VeriCheck ddPCR *Mycoplasma* Detection Kit does not detect DNA of three species of bacteria (*S. bovis*, *L. acidophilus*, *C. sporogenes*) as described by the European Pharmacopoeia

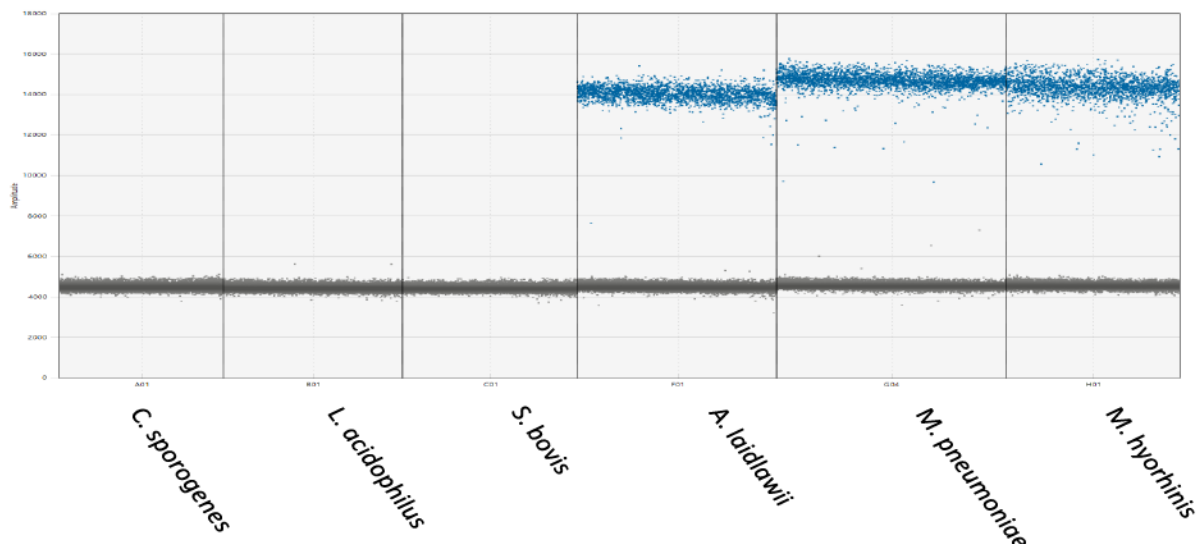


Figure 1. 1D plot of specificity test showing no detection of *S. bovis*, *L. acidophilus* and *C. sporogenes* at comparable input GC as the three representative *Mycoplasma* species.

Inhibitor Tolerance: Kit has no inhibition in cp/μl for SDS, EDTA, and DMSO

Each inhibitor condition was done in four replicates and compared to the no inhibitor control sample ($n = 3$). The mean cp/μl of both the positive control plasmid and internal control in the presence of each inhibitor was compared to those of the no inhibitor control sample to calculate percentage inhibition. The results demonstrate that the VeriCheck ddPCR *Mycoplasma* Detection kit has no inhibition in cp/μl for any of the inhibitors tested at the specified levels. Up to 0.01% SDS, 1.25mM EDTA, 1% CHO, HEK293, and HEK293T lysates, up to 10% media and 5% DMSO experienced no inhibition or changes to cp/μl readouts (figure 2).

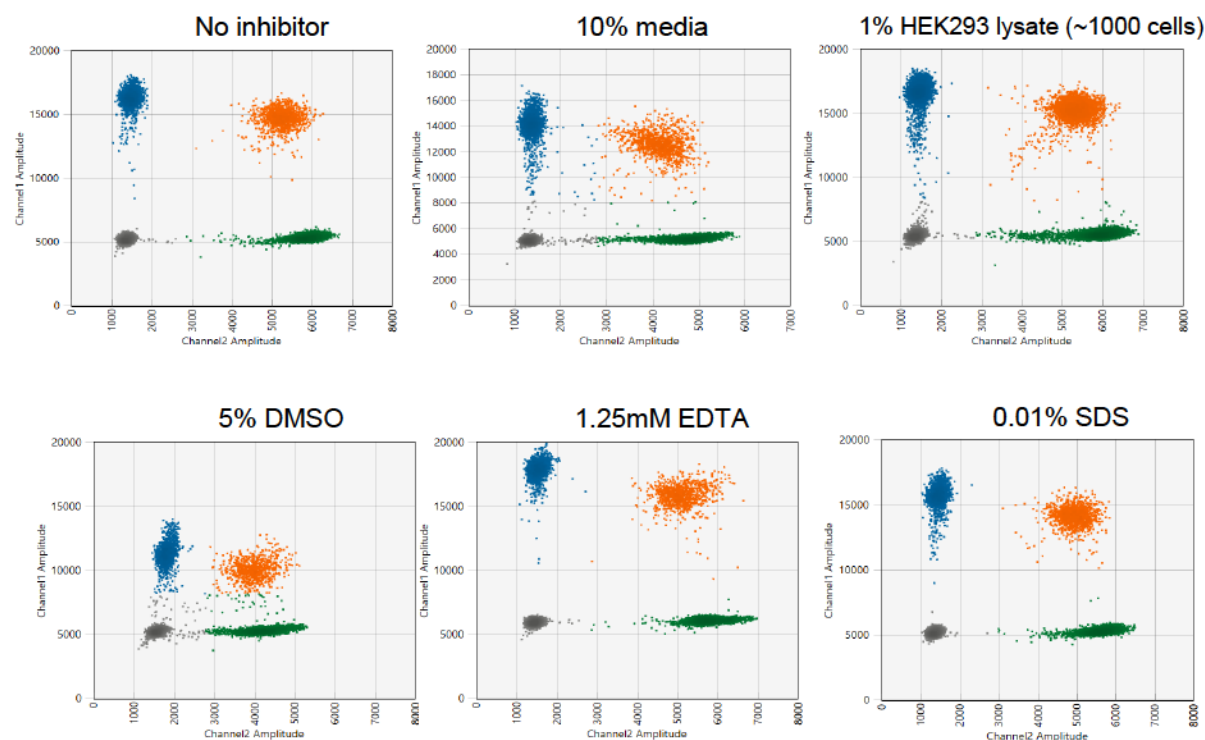


Figure 2. 2D plots of 5000cp/well of *Mycoplasma* gDNA in the presence of inhibitors.

